

Notes: Broer, Hansen, Krusell & Öberg (RES, 2020)

The New Keynesian Transmission Mechanism: A Heterogeneous-Agent Perspective

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1 Big picture

- **Question.** How does monetary policy transmit to aggregate activity when households are heterogeneous, incomplete markets prevent full insurance, and firm ownership is concentrated among capitalists?
- **Core contribution.** The paper builds a tractable heterogeneous-agent New Keynesian model whose log-linear reduced form is almost as simple as the textbook representative-agent model.
- **Central contrast.** In RANK, the source of nominal rigidity does not qualitatively change the output response. In this HANK model, the source of nominal rigidity is decisive.
- **Rigid-prices/flexible-wages result.** Monetary policy redistributes resources between workers and capitalists, but does not move hours or output.
- **Rigid-wage result.** Wage rigidity restores standard monetary transmission: labor becomes demand-determined and output responds to monetary shocks.
- **Economic takeaway.** Heterogeneity changes which nominal friction matters. Sticky-price-only RANK can generate plausible aggregate responses for the wrong microeconomic reason.

2 Model objects and measurement

2.1 Workers and capitalists

There is a unit mass of workers and a small mass of capitalists. Workers supply labor, receive wages, face idiosyncratic labor productivity shocks, and can trade only a risk-free bond subject to a non-borrowing constraint. Capitalists own firms, receive dividends, and do not work.

Interpretation. The key heterogeneity is a stark ownership structure: workers receive wages, capitalists receive profits.

2.2 Incomplete markets

Workers can smooth idiosyncratic shocks only through non-contingent bonds. The borrowing constraint is tight and bond assets are in zero net supply. This makes worker consumption effectively autarkic and analytically tractable.

Interpretation. The model keeps incomplete-market intuition but removes high-dimensional wealth-distribution dynamics.

2.3 Nominal rigidities

The baseline has sticky prices and flexible wages. Section 3 adds wage adjustment costs, producing a wage Phillips curve and a wage-accounting equation.

Interpretation. The comparison between flexible and rigid wages is the central theoretical experiment.

3 Models and theoretical specifications

3.1 Worker problem

$$\max E_0 \sum_t \beta^t \left[\log C_{jt} - \frac{N_{jt}^{1+\varphi}}{1+\varphi} - \vartheta \mathbf{1}\{N_{jt} > 0\} \right]$$

subject to

$$P_t C_{jt} + Q_t B_{jt} = W_{jt} N_{jt} + B_{j,t-1}, \quad B_{jt} \geq 0.$$

Interpretation. Balanced-growth preferences imply that when worker consumption equals labor income, wage changes have offsetting income and substitution effects on hours.

3.2 Flexible-wage HANK system

$$\begin{aligned} \pi_t &= \beta E_t \pi_{t+1} + \lambda_p \hat{\omega}_t, \\ \hat{c}_t &= E_t \hat{c}_{t+1} - (\hat{i}_t - E_t \pi_{t+1}), \quad \hat{i}_t = \phi_\pi \pi_t + \nu_t, \\ \hat{c}_t &= \hat{\omega}_t + \hat{n}_t, \quad \hat{\omega}_t = \hat{c}_t + \varphi \hat{n}_t. \end{aligned}$$

Combining market clearing and labor supply gives $\hat{n}_t = 0$.

Interpretation. Monetary policy moves wages, profits, inflation, and consumption distribution, but not hours or output.

3.3 Textbook RANK comparison

$$\hat{c}_t = \bar{S}(\hat{\omega}_t + \hat{n}_t) + (1 - \bar{S})\hat{d}_t.$$

Interpretation. RANK households consume both wages and dividends. This profit-income effect is what generates the textbook labor-supply response.

3.4 Rigid-wage HANK system

$$\begin{aligned} \pi_t^p &= \beta E_t \pi_{t+1}^p + \lambda_p \hat{\omega}_t, \\ \pi_t^w &= \beta E_t \pi_{t+1}^w - \lambda_w [\hat{\omega}_t - (\hat{c}_t + \varphi \hat{n}_t)], \\ \hat{\omega}_t &= \hat{\omega}_{t-1} + \pi_t^w - \pi_t^p, \\ \hat{c}_t &= E_t \hat{c}_{t+1} - (\hat{i}_t - E_t \pi_{t+1}^p), \quad \hat{i}_t = \phi_\pi \pi_t^p + \nu_t, \quad \hat{c}_t = \hat{\omega}_t + \hat{n}_t. \end{aligned}$$

Interpretation. Wage rigidity creates a wedge between the real wage and workers' marginal rate of substitution. Labor becomes demand-determined and output responds.

4 Theoretical design and identification logic

The paper is not an empirical causal design. It is a mechanism-identification exercise. The authors hold the firm side close to the textbook NK model and change the household ownership structure. Comparing RANK and HANK under the same monetary shock isolates the representative-agent profit-income channel. The tight borrowing constraint is a tractability device, while the wage-rigidity extension tests whether monetary transmission survives once workers cannot freely adjust wages and hours.

5 Section-by-section study guide

- **Introduction.** Motivates why heterogeneous income sources matter for monetary transmission.
- **Section 2.** Builds the sticky-price/flexible-wage HANK model and shows that monetary shocks redistribute but do not move output.
- **Section 2.6.** Figure 1 compares RANK and HANK impulse responses.
- **Section 3.** Adds wage rigidity and derives the wage Phillips curve.
- **Section 3.5.** Figure 2 shows that rigid wages restore aggregate monetary transmission.
- **Conclusion.** Argues that sticky-price-only RANK is a problematic benchmark and that a simple sticky-wage HANK model is preferable.

6 Tables and figures

6.1 Table 1: Parameter values used for computing the impulse-response functions

Read: The calibration uses $\beta = 0.99$, $\varphi = 1$, $\epsilon_p = 6$, $\theta_p = 2/3$, $\phi_\pi = 1.5$, and $\rho_\nu = 1/2$.

Detailed interpretation. The table anchors the comparison in standard textbook parameters. The surprising findings therefore come from household heterogeneity and the source of nominal rigidity, not unusual calibration.

Economic message. The table makes the mechanism critique sharper: a familiar calibration produces no output response in HANK when wages are flexible.

TABLE 1
Parameter values used for computing the impulse-response functions (IRFs)

Parameter		Value
Discount factor	β	0.99
Frisch elasticity	φ	1
Elast. of substitution	ϵ_p	6
1-reset probability	θ_p	2/3
Taylor coefficient	ϕ_π	1.5
Persistence of monetary policy shock	ρ_ν	1/2

Parameter values used for computing the impulse-response functions.

6.2 Figure 1: Equilibrium responses under rigid prices and flexible wages

Read: The textbook RANK model is on the left and the simple HANK model on the right. Both models show a real-rate increase, lower inflation, lower real wages, and higher profits. But only RANK shows lower hours and output; HANK hours and output are flat.

Detailed interpretation. In RANK, the representative household receives profits, so higher profits after a tightening create a wealth effect that lowers labor supply. In HANK, profits go to capitalists, not workers. Worker income and substitution effects cancel, so the real wage absorbs the shock and hours do not move.

Economic message. Sticky-price-only RANK generates a plausible aggregate contraction through an implausible representative-household profit-income channel.

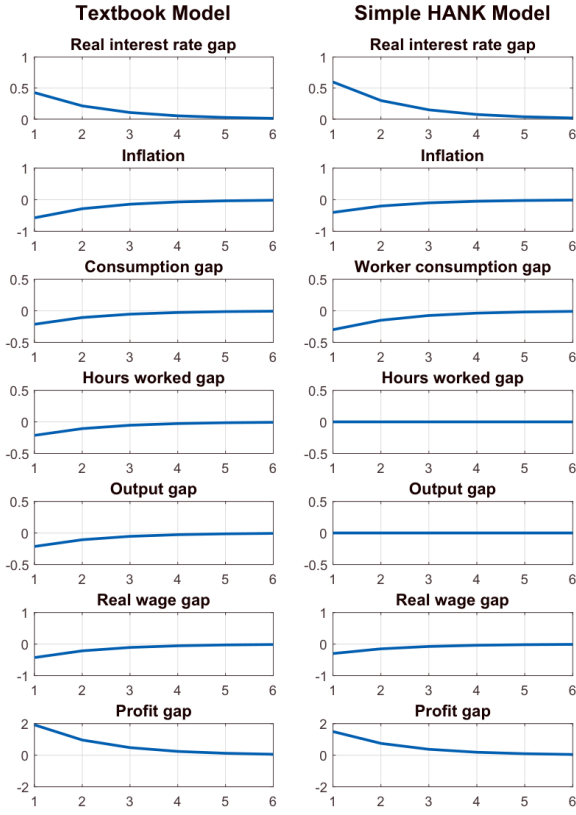
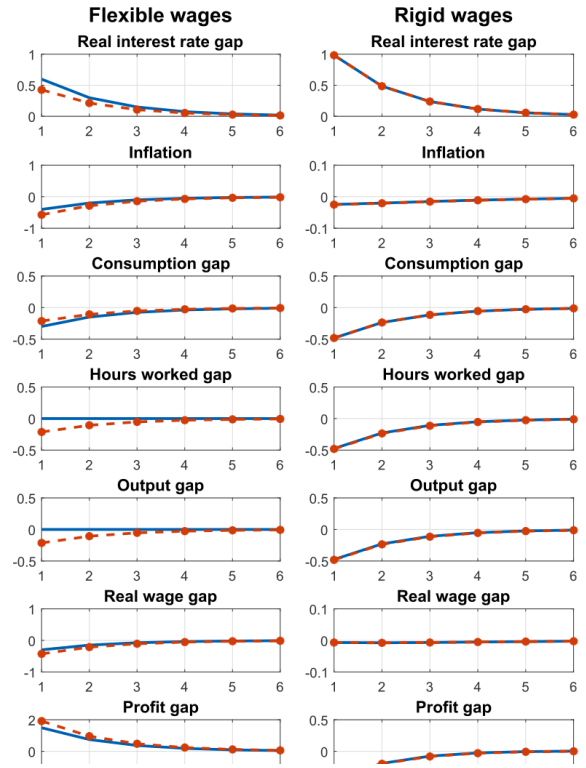


FIGURE 1

Equilibrium responses, measured as percentage deviations from the steady state, to a positive twenty-five basis-point shock in the policy rate. Textbook RANK model (left); Our simple HANK model (right). Inflation and interest rates:



6.3 Figure 2: Equilibrium responses with flexible wages versus rigid wages

Read: Flexible wages are shown on the left; rigid wages on the right. Solid lines are HANK, dashed lines are RANK. Under flexible wages, HANK output and hours are flat. Under rigid wages, HANK and RANK look much more similar and output falls after the tightening.

Detailed interpretation. Wage rigidity prevents the real wage from doing all the adjustment. Workers must meet labor demand, so hours and output move. Factor shares are more stable and distributional effects are smaller.

Economic message. Wage rigidity, not price rigidity alone, is central for monetary transmission once workers and profit recipients are distinct groups.

7 Result map

- **Analytical HANK benchmark.** The model keeps incomplete-market heterogeneity but becomes a small linear system.
- **No output response with flexible wages.** Monetary policy redistributes income but does not affect hours or output.
- **RANK profit-income channel.** The textbook model's output response relies on the representative worker receiving profits.
- **Wage rigidity restores transmission.** Sticky wages create a wedge between wages and the marginal rate of substitution, making labor demand move output.
- **Nominal friction source matters.** Price and wage stickiness are not interchangeable in HANK.

8 Short summary

- The paper studies monetary transmission in a tractable HANK model.
- Workers receive wages and face idiosyncratic risk; capitalists receive profits.
- With sticky prices and flexible wages, monetary policy has distributional effects but no output effects.
- The textbook RANK contraction relies on a profit-income wealth effect for the representative household.
- Adding wage rigidity restores the standard output and hours response.
- The key lesson is that heterogeneity makes the source of nominal rigidity first-order.

9 Conclusion

9.1 Core conclusion

Broer et al. show that the textbook New Keynesian transmission mechanism is not robust to a simple heterogeneous-agent structure. If workers do not receive profits and wages are flexible, monetary policy affects distribution but not output.

9.2 Economic intuition

Flexible wages keep workers on their labor-supply curve. With balanced-growth preferences and worker consumption equal to labor income, wage changes have offsetting income and substitution effects. RANK breaks this cancellation because the same household receives profits.

9.3 Methodological contribution

The paper provides a low-dimensional HANK benchmark that exposes exactly where representative-agent aggregation changes monetary transmission.

9.4 Policy implication

Policy models should take seriously who receives profits, who supplies labor, and whether wages are rigid. Sticky prices alone may be an insufficient benchmark.

9.5 Limitations

The model abstracts from capital, habits, unemployment, and richer wealth distributions. Its tractability relies on an extreme no-risk-sharing limit.

9.6 Takeaway

Monetary transmission is jointly about nominal rigidities and income ownership. A model can fit aggregate responses while using an implausible distributional mechanism.